

BLP v1 — TSL-ONE-S Multimodal Sign Language → Speech

Blueprint, implementation and results.

Run date: 2026-09-06 • Environment: conda `hugging` (Python 3.10.16, PyTorch 2.7.0+cu128) • GPU: NVIDIA RTX 2050 (4.3 GB)

Deliverables: [slm_labs.ipynb](#) (executed, 60 cells, 14 figures, 0 errors) • [modules/](#) • `artifacts/mssf_best.pt`

1 • What the data actually is

Before designing anything, the released bundle was inspected rather than assumed.

Property	Finding
Files	4 152 <code>.npy</code> clips in <code>data/TSL-ONE-Pose/</code>
Filename grammar	<code><signer>_<category>_<gloss>.npy</code> — 29 signers × 8 categories × 184 gloss classes
Array shape	<code>(T, 150)</code> → reshaped <code>(T, 75, 2)</code> ; T ranges 16–194, median 54; 240 554 frames total
Landmark layout	MediaPipe Holistic: <code>0–32</code> pose (33), <code>33–53</code> left hand (21), <code>54–74</code> right hand (21)
Coordinates	normalised image space, x/y only (no z, no visibility)
Missing data	exact <code>(0, 0)</code> = "not detected": 8.5 % of frames for the left hand, 7.1 % for the right
Samples per gloss	17–27
Signers per gloss	≥ 17 for every gloss

Correction to the brief. The task description says *"54 landmarks ... and a neck landmark for SPOTER"*. The

released `.npy` does **not** carry the 54-point SPOTER subset — it carries the raw 75-point MediaPipe Holistic set (33 + 21 + 21). Landmark identity was verified empirically from mean positions (nose at the top-centre, shoulders symmetric about it, hand blocks tracking their pose wrists) and from the zero-rate pattern, which matches the hand blocks exactly. The neck is not a landmark; it is derived here as the shoulder midpoint. Pose landmarks 23–32 (legs) are never zero but are MediaPipe extrapolations for an upper-body recording and carry no signal, so they are dropped.

Consequence for "Face Impression". There is **no Face Mesh in the release**.

The only facial evidence is pose landmarks 0–10 : nose, six eye points, two ears, two mouth corners. That is enough for head pose, head motion, gaze line and mouth-width dynamics — a *coarse* face-impression channel — but **not** for brow, eyelid or cheek articulation. The face stream was therefore built and fused exactly as a dense face stream would be, so Face Mesh is a drop-in upgrade, and its real contribution was **measured by ablation** rather than asserted (§ 6).

2 • Blueprint

2.1 Architecture — MSSF-Net (Multi-Stream Sign Fusion Network)



Design decisions and the reason for each

Decision	Why
Per-stream encoders before fusion	hands, body and face differ in dimensionality, noise and missing-data behaviour; a shared encoder would have to average over all three

Hand <i>shape</i> wrist-local & scale-free, hand <i>place</i> kept in body coords	in TSL, handshape and location are independent phonological parameters — collapsing them loses one
Learned <code>tanh(g)</code> gate on the face channel	lets the model shut a weak modality off instead of being forced to inject noise; the gate value is then readable evidence
Transformer over GRU for the temporal trunk	a sign is a stroke bracketed by preparation/retraction; attention can down-weight the brackets, recurrence cannot as cleanly
Attention pooling, not mean pooling	same reason — measured in § 6, the model concentrates on the middle of the clip
ArcFace embedding head alongside softmax	angular margin compacts class clusters, which is what makes prototype distance a usable open-set score
64-frame resampling	median clip is 54 frames; fixed length makes every batch a dense tensor with no padding mask overhead

2.2 Feature engineering

One vectorised pass per clip (`modules/feature_engineering.py`):

1. **Drop-out repair** — `(0,0)` → `NaN` → linear interpolation along time; a per-frame **presence flag** is kept as an explicit feature so the model knows the difference between "hand at rest" and "hand not seen".
2. **Body frame** — origin at the neck (shoulder midpoint), scale = median shoulder width ⇒ translation- and size-invariant.
3. **Hand streams** — 21 wrist-local scale-free points + wrist position in body coords + 5 fingertip-to-wrist distances + finger span.
4. **Pose stream** — 12 upper-body points in body coords + 5 relational distances (inter-hand, each hand↔face, each hand↔neck).
5. **Face stream** — 11 nose-centred, inter-ocular-scaled points + 8 geometric descriptors (mouth width, jaw-drop proxy, ear-span/yaw proxy, eye-mouth/pitch proxy, head roll cos/sin, head shift x/y).
6. **Dynamics** — `ΔX` and `Δ²X` appended to every stream.
7. **Uniform time** — linear resample to 64 frames.

Resulting per-frame dimensionality: **LH 151 • RH 151 • Pose 87 • Face 60 = 449.**

2.3 Leakage protocol

A random split is not acceptable here: each signer repeats each gloss, so random shuffling puts near-duplicate performances of the same person on both sides.

- **Signer-independent split** — test = signers 05, 12, 24 ; val = 06, 20 ; train = the other 24. Sizes **3 302 / 298 / 552**.
- Test signers were chosen among those with full 184-gloss coverage, so the test set exercises the entire vocabulary.
- **Audited in the notebook (§ 5)** on: signer overlap (all three pairs), index duplication, dataset coverage, filename overlap, unseen-gloss check, and a **SHA-1 content hash of every raw test clip against every train clip**. The cell asserts and halts on failure.
- The standardiser (per-feature mean/std) is fitted on **train rows only**. Early stopping uses validation. The open-set threshold is calibrated on **validation only**.

Audit result: PASS — 0 signer overlaps, 0 duplicate indices, 0 filename overlaps, 0 content-duplicate test clips, full dataset coverage, 184/184 glosses present in each partition.

2.4 Optimisation for scale

Concern	Treatment
Disk I/O in the training loop	none — the whole corpus is materialised once as contiguous <code>float32 (N,64,D)</code> tensors (477 MB) and sliced by index
Per-item Python work	none — no <code>Dataset.__getitem__</code> , no DataLoader workers; a batch is one <code>index_select</code> plus one pinned host→device copy
Augmentation cost	fully batched on the GPU : tempo warp by gathered linear interpolation, amplitude jitter, Gaussian noise, frame dropout, hand-dropout, all as tensor ops
Feature pass	vectorised — sliding mean via cumulative sum, resampling via a single two-index gather-and-blend (no <code>np.interp</code> per column). 4 152 clips in ~16 s
Numerics	mixed precision (AMP) + gradient clipping; cosine LR with warm-up

Throughput: **~4.0 s/epoch** for MSSF-Net (52 batches), ~3.0 s/epoch for the baseline.

3 • Model selection (validation only)

A sweep was run first; the winner was picked on **validation macro-F1**, never on test.

run	layers	dropout	mixup α	lr	val macro-F1
MSSF-Net A	4	0.20	0.2	3e-4	0.8721
MSSF-Net B (deeper)	6	0.25	0.2	3e-4	0.8591
MSSF-Net C — selected	4	0.20	0.4	4e-4	0.8895

Depth beyond 4 layers over-fits 3.3 k training clips; stronger mixup with a slightly higher LR is what actually helps. The selected recipe was then used for **every** model in the notebook so all comparisons are like-for-like.

4 • Results — sign recognition (test = 3 unseen signers, 552 clips)

model	params	accuracy	top-5	macro-F1	balanced acc.	ECE
Baseline BiGRU	2.28 M	0.8859	0.9783	0.8797	0.8859	0.2038
MSSF-Net, face zeroed (retrained)	4.05 M	0.9203	0.9801	0.9182	0.9203	0.1079
MSSF-Net (final)	4.05 M	0.9293	0.9728	0.9281	0.9293	0.1147

- **+4.8 macro-F1 points over the baseline**, on signers the model has never seen.
- Calibration also improves markedly (ECE 0.204 → 0.115), i.e. the confidences are more trustworthy, which is what makes the rejection stage possible.
- 39 / 552 clips misclassified. The residual errors are concentrated in glosses with only 3 test samples and in a handful of visually adjacent pairs.

5 • Results — open-set detection

Scoring a closed-set model on its own test set cannot measure unknown-sign behaviour, so a **probe model** was trained on 160 glosses with **24 glosses held out entirely**; those 24 are genuine unknowns, produced by unseen signers.

rejection score	AUROC	AUPR (unknown)
max-softmax	0.9346	0.6985
prototype cosine	0.9322	0.7015
prototype margin	0.9154	0.5953
fused (deployed)	0.9346	0.7040

- **OSCR-AUC = 0.8758**
- At the validation-calibrated operating point (90 % known-acceptance target): **92.5 % of known signs accepted, 66.7 % of novel signs correctly rejected.**
- Applied to the production 184-gloss model, rejection turns 92.93 % closed-set accuracy into **89.86 % accuracy with a _ escape hatch**: 31 clips returned _, and **14 of those 31 would have been wrong answers anyway**. The system prefers _ to a confident lie.

6 • Results — does the face channel help?

Three independent pieces of evidence:

1. **Signal test.** Between-gloss variance ratio (η^2) of the eight face descriptors: `roll_cos` 0.427, `head_dx` 0.172, `eye_mouth` 0.157, `jaw_drop` 0.133, `roll_sin` 0.130, `head_dy` 0.117, `ear_span` 0.076, `mouth_width` 0.048. Head orientation and head translation carry the most gloss-discriminative information; mouth width carries the least — consistent with only having two mouth corners.
2. **Ablation retrain.** Identical architecture, face stream zeroed, same seed and budget: **0.9203 → 0.9293 accuracy (+0.90 pt), 0.9182 → 0.9281 macro-F1 (+0.99 pt).**
3. **Learned gate.** `tanh(g)` = +0.060 — small but clearly non-zero; the model keeps the channel rather than switching it off.

Verdict: the coarse face channel is a real but modest contributor at this landmark resolution. It is *not* facial-expression recognition, and the report

does not claim it is. The headroom is in re-extracting TSL-ONE-S with MediaPipe Face Mesh; the architecture is already wired for it.

7 • Results — sequence and translation stages

TSL-ONE-S is an **isolated-sign** corpus: no sentence annotations, no paired Thai translations. To exercise and measure the sequence stage honestly, 80 continuous utterances (3–6 signs each) were **composed from held-out test clips** — every clip is still one the model has never seen, from a signer it has never seen — and read left-to-right by the recogniser with rejection enabled.

stage	metric	value
Sign sequence	WER	0.0950
Sign sequence	CER	0.0627
Sign sequence	sentence exact match	0.5875
Gloss-string output	BLEU (char-tokenised)	93.47
Gloss-string output	chrF	93.26

BLEU/chrF use character tokenisation, which is the correct setting for Thai (no orthographic word boundaries). **These score gloss-string fidelity, not Thai fluency** — a true SLT BLEU needs paired Thai sentences the corpus does not ship.

8 • Results — inference

Route 1 — held-out test set (552 clips, 3 unseen signers). Closed-set accuracy 0.9293; with rejection on, 0.8986 with 31 _ outputs. The notebook renders a visual panel per clip: mid-stroke skeleton, predicted gloss coloured by correctness, and the top-5 posterior.

Route 2 — `data_test/` . The folder was empty at run time (0 files). Nothing new could be scored. The route is implemented and ready in both formats:

- `.npy` of shape `(T,150)` or `(T,75,2)` → `INF.load_npy_clip()`

- raw video `.mp4/.mov/.avi/.webm/.mkv` → `INF.extract_from_video()` , which runs the MediaPipe **Tasks** API (mediapipe 1.0.1 no longer ships the legacy `mp.solutions` graphs) and emits the identical 75-landmark ordering, so the same feature pass applies unchanged.

Drop files into `data_test/` and re-run that single cell — nothing else changes.

Route 3 — sequence → sentence → speech. The recogniser emits **gloss ids**, because the release contains no Thai lemma table. The sentence stage therefore **declines to call the LLM on opaque ids** and passes them through — asking a language model to "translate" `GLOSS_0272` produces a confident hallucination, which would be worse than no output:

```
recognised gloss sequence : 0272 0282 0207 0240 _ 0007
reference gloss sequence  : 0272 0282 0207 0240 0922 0007
sentence stage (no-lexicon: LLM skipped, gloss ids carry no lexical
content)
  GLOSS_0272 GLOSS_0282 GLOSS_0207 GLOSS_0240 _ GLOSS_0007
```

The stage is then demonstrated end to end with an explicitly illustrative lexicon, which shows the LLM doing the job gloss order requires — adding particles and Thai word order — while **preserving _ instead of guessing**:

```
gloss sequence (illustrative) : อนุ อนุอนุ _ อนุอนุอนุ อนุอนุอนุอนุอนุ
LLM composition (gpt-4o-mini) : อนุอนุอนุ _ อนุอนุอนุอนุอนุอนุอนุอนุอนุ
TTS (gpt-4o-mini-tts)         : artifacts/inference_demo.mp3
```

Supplying the real lemma table via `lexicon=` is the only change needed to make that the live path for recognised signs.

9 • Consolidated evaluation

Stage	Metric	Value
Sign recognition (closed-set, unseen signers)	Accuracy	0.9293

Sign recognition (closed-set, unseen signers)	Macro-F1	0.9281
Sign recognition (closed-set, unseen signers)	Top-5	0.9728
Sign recognition (open-set, rejection on)	Accuracy	0.8986
Unknown detection (160 known / 24 novel)	AUROC	0.9346
Unknown detection (160 known / 24 novel)	OSCR-AUC	0.8758
Unknown detection (160 known / 24 novel)	Unknown recall @ 90 % known-accept	0.6667
Sign sequence	WER	0.0950
Sign sequence	CER	0.0627
Gloss-string output	BLEU (char)	93.47
Gloss-string output	chrF	93.26
Metric-space quality	prototype nearest-neighbour accuracy	0.9221
Metric-space quality	cosine to own prototype vs. nearest impostor	0.767 vs 0.385

Total notebook wall-clock: **17.7 min** end to end on one RTX 2050 (data load 2 s, feature engineering 16 s, four model trainings, all evaluation and inference).

10 • What was built

modules/	
config.py	paths, landmark topology, split definition,
all hyperparameters	
data_io.py	indexing, loading, signer-independent split,
leakage audit	
feature_engineering.py	vectorised raw → 4-stream tensors
dataset.py	in-RAM StreamBank + batched GPU augmentation +
mixup	
models.py	BaselineGRU, MSSFNet, ArcMarginHead, soft-
target CE	
train.py	AMP training loop, cosine+warm-up, early
stopping	
evaluate.py	recognition / sequence (WER, CER) /
translation (BLEU, chrF) / open-set metrics	
openset.py	prototypes, fused rejection score, threshold
calibration, OSCR	
inference.py	SignPredictor, video→landmarks, LLM
composition, TTS	
viz.py	Thai-capable global font, skeletons, heatmaps,
curves	
slm_labs.ipynb	16 sections, executed end to end, 14 figures,
0 errors	
artifacts/	
mssf_best.pt	the one production checkpoint (17.1 MB) –
weights, standardiser,	
	prototypes, fusion stats, calibrated
threshold, class names	
inference_demo.mp3	TTS output of the demonstration sentence

Fonts: `font/THSarabunNew*.ttf` registered as the global matplotlib family, so every figure renders Thai and Latin. LLM and TTS credentials are read from `.env` and never printed.

11 • Honest limitations

1. **data_test/ was empty**, so the "new real-world data" requirement could not be exercised on real material. The code path exists, is typed and is one cell away from running.
2. **No Thai lemma table**. Gloss ids stay ids. BLEU/chrF therefore measure gloss-string fidelity, and the LLM sentence in the notebook is an explicitly-labelled demonstration, not a scored translation result.
3. **No dense facial landmarks**. "Face impression" is head-pose-and-mouth-corner level, not expression recognition. The measured contribution (+0.9 accuracy points) should be read with that in mind.
4. **Isolated signs only**. WER/CER are computed on utterances composed from held-out isolated clips. This measures the recogniser under sequence-level scoring; it does not measure continuous-signing segmentation, which the corpus cannot support.
5. **Small test set per class**. 552 test clips over 184 glosses is ~3 per class, so per-gloss F1 is coarse; the aggregate numbers are the reliable ones.
6. **One split**. Results are for one signer-independent partition, not a cross-validated mean over signer folds.

12 • Highest-value next steps

1. Re-extract TSL-ONE-S with **MediaPipe Face Mesh** (468 points) and re-run § 10 of the notebook — the fusion path is already in place and this is where the remaining headroom is.
2. Add the **Thai lemma table** to `lexicon` — this activates the LLM and TTS stages for real and makes BLEU/chrF genuine translation metrics.
3. **Leave-one-signer-out cross-validation** over all 29 signers for a variance estimate instead of a single split.
4. **Continuous signing**: sliding-window inference plus CTC over the existing trunk, which would make WER/CER measure real segmentation.
5. Cheap accuracy wins that were not applied here: test-time augmentation and a small seed ensemble.